Winning at Failure!

Modern Management of Cardiogenic Pulmonary Edema

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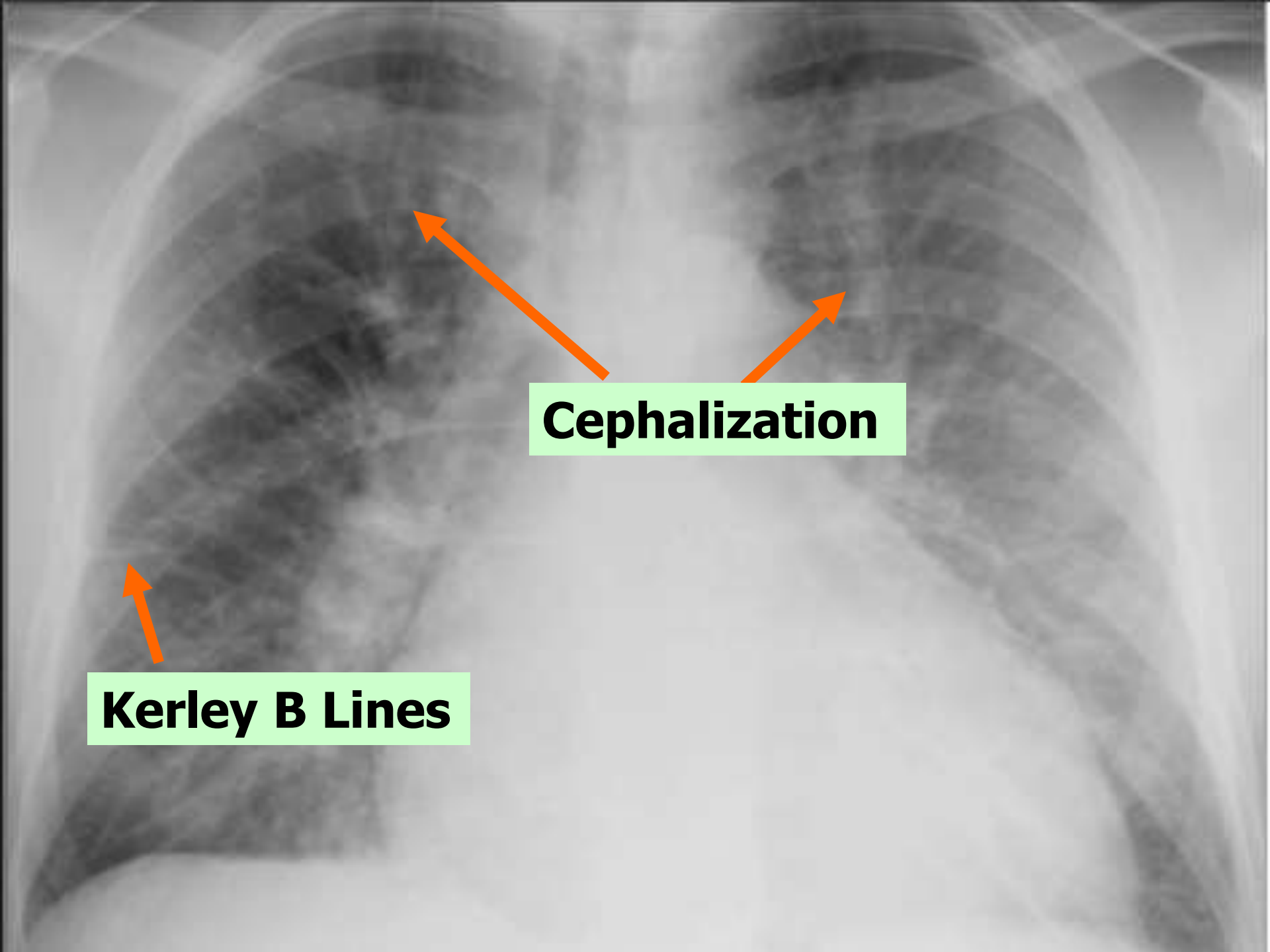
Lectures.umem.org/SEMA

Case

- 65 yo. man presents with dyspnea
  - Developed over the past 6 hours
  - History of hypertension, diabetes
  - Diaphoretic, normal mental status
  - Afebrile, HR 110, BP 180/110, RR 26, pox 88%
  - Lungs — crackles, JVD


Cephalization
Kerley B Lines

Case

- 100% NRB — pulse ox 93%
- Monitor, IV, ECG
  - Sinus tachycardia, no acute ST/T-wave abnormalities

Case

What is the optimal treatment in the first 5-10 minutes??

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OUTLINE

- Definition
- Pathophysiology
- Pharmacological management
  - Preload reduction
  - Afterload reduction
  - Inotropic support
- Noninvasive Positive Pressure Ventilation
- Prehospital Issues
- Summary

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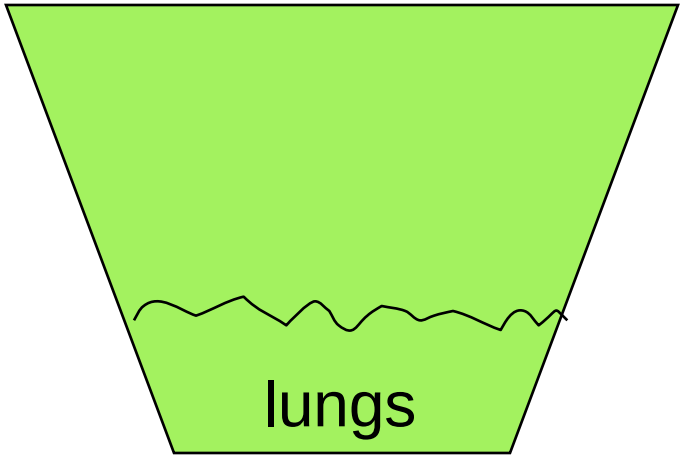
CARDIOGENIC PULMONARY EDEMA

- Definition:
  - Leakage of fluid from the pulmonary capillaries and venules into the alveolar space as a result of increased hydrostatic pressure
  - Inability of the LV to effectively handle its pulmonary venous return

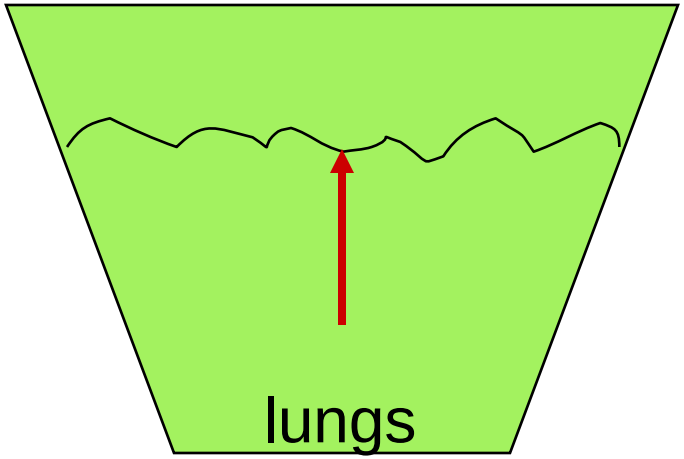
ANATOMY

Normal

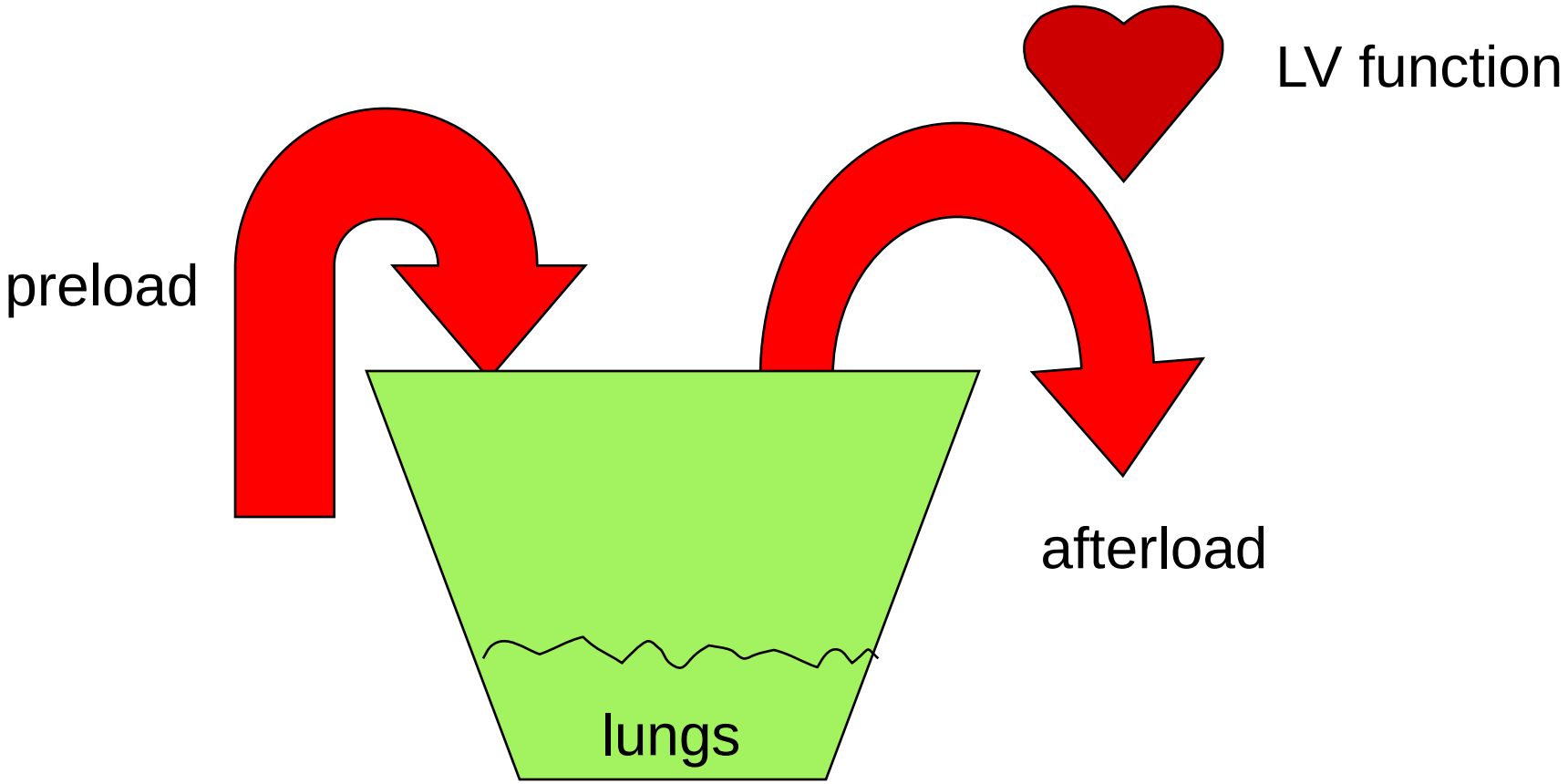


PATHOLOGY

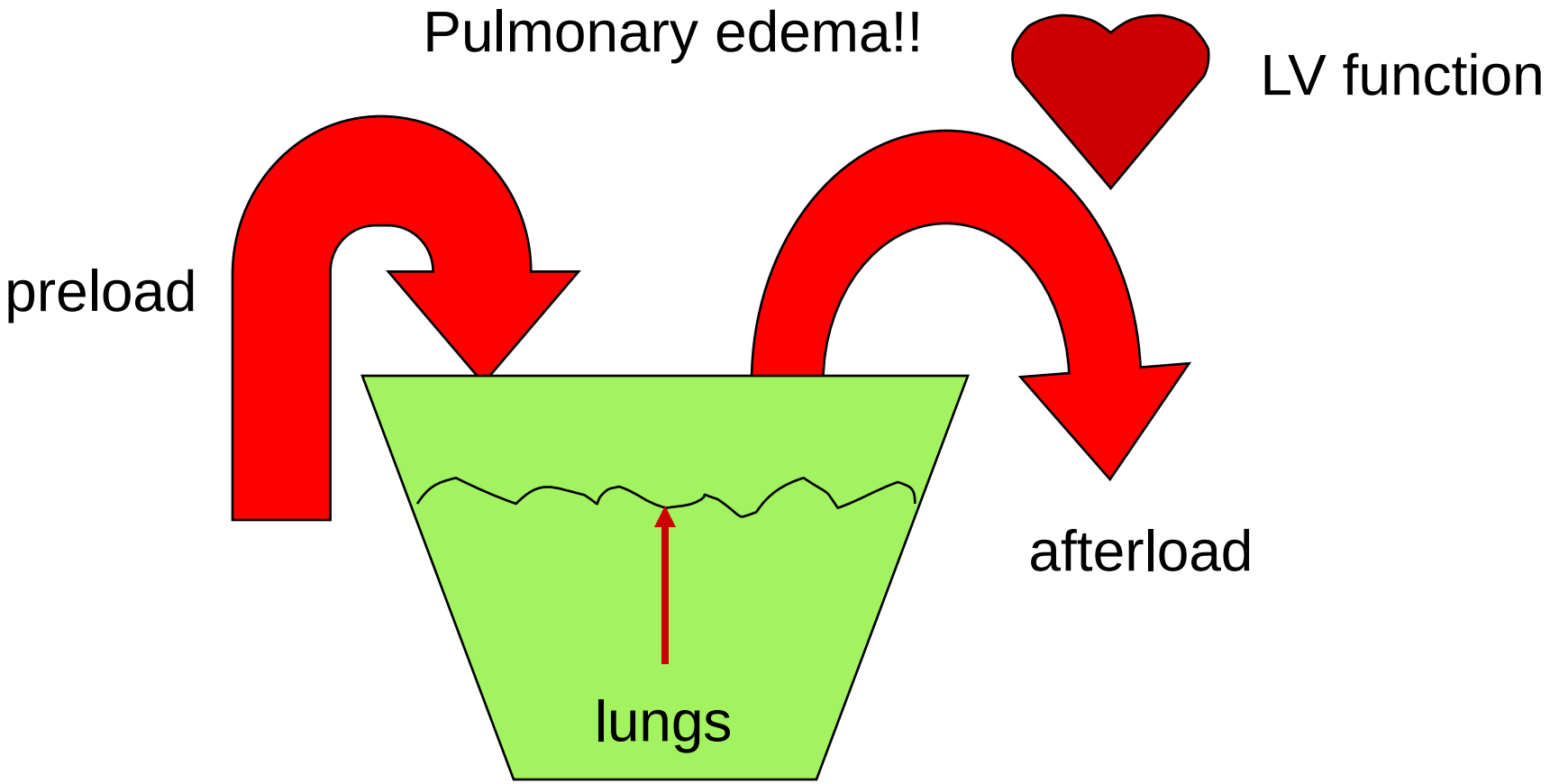
Pulmonary edema!!



PHYSIOLOGY



PATHOPHYSIOLOGY



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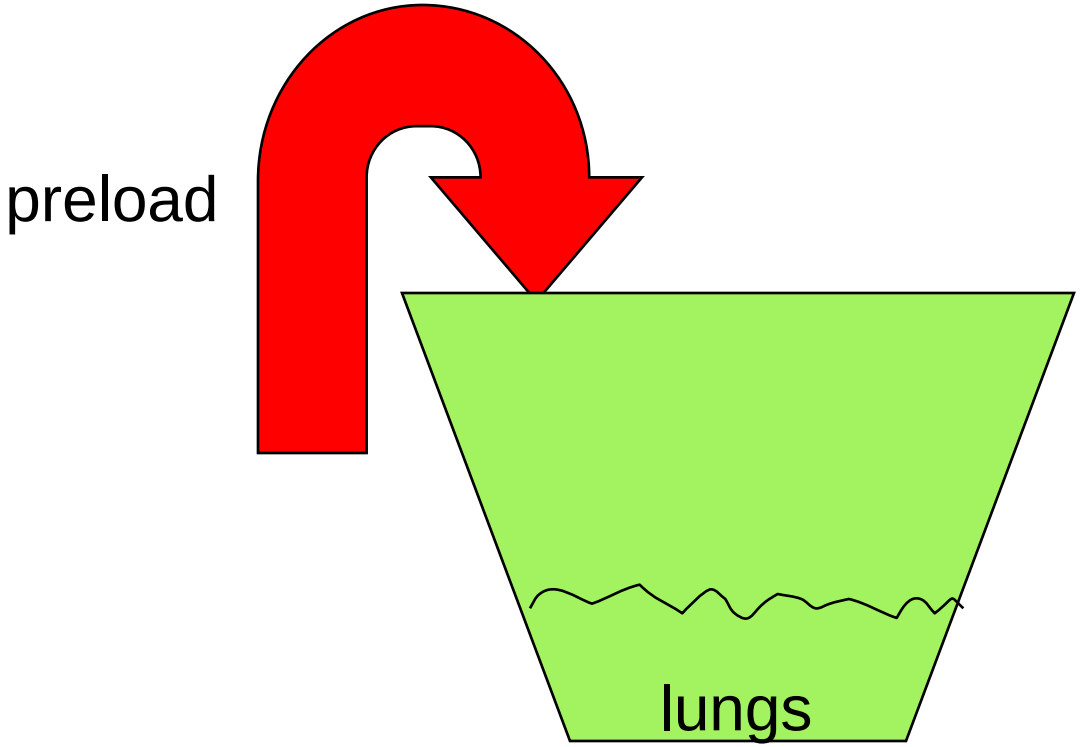

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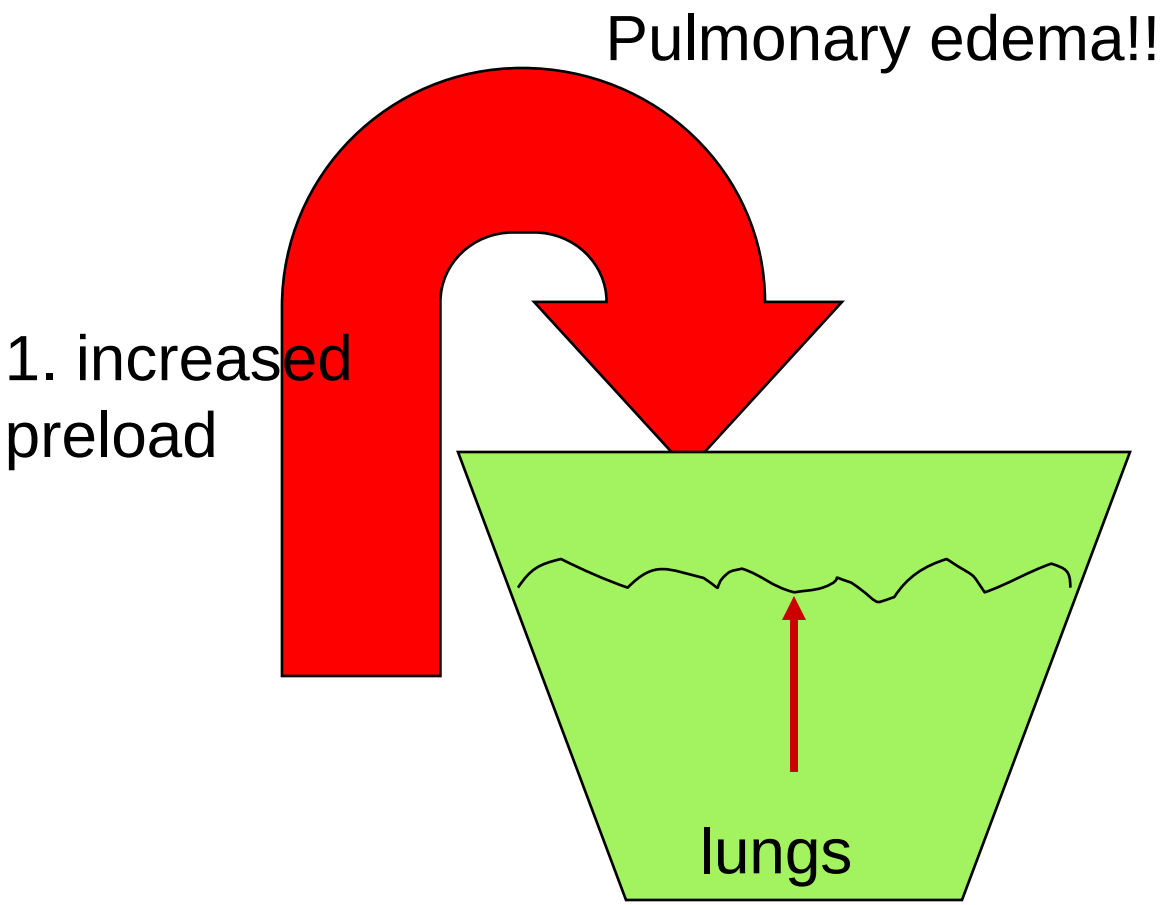

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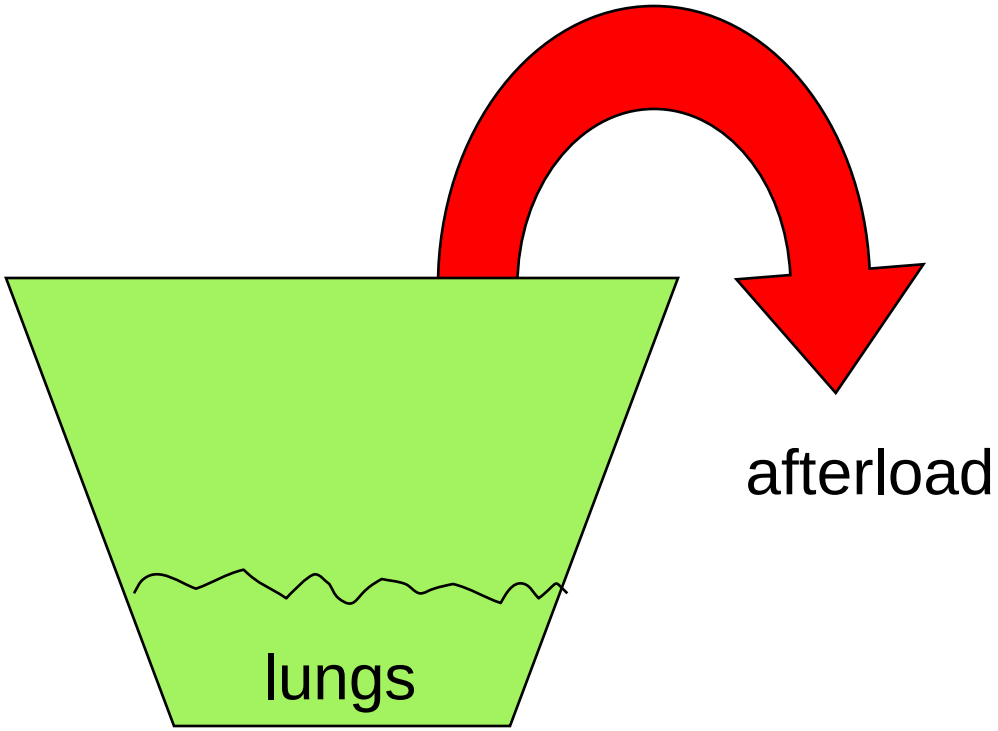

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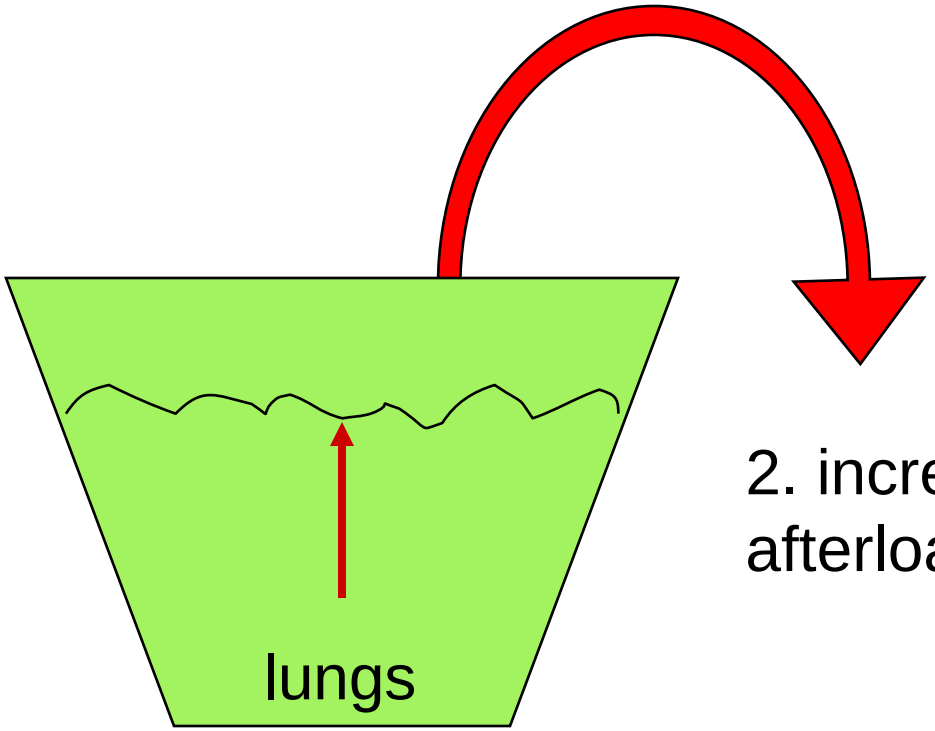
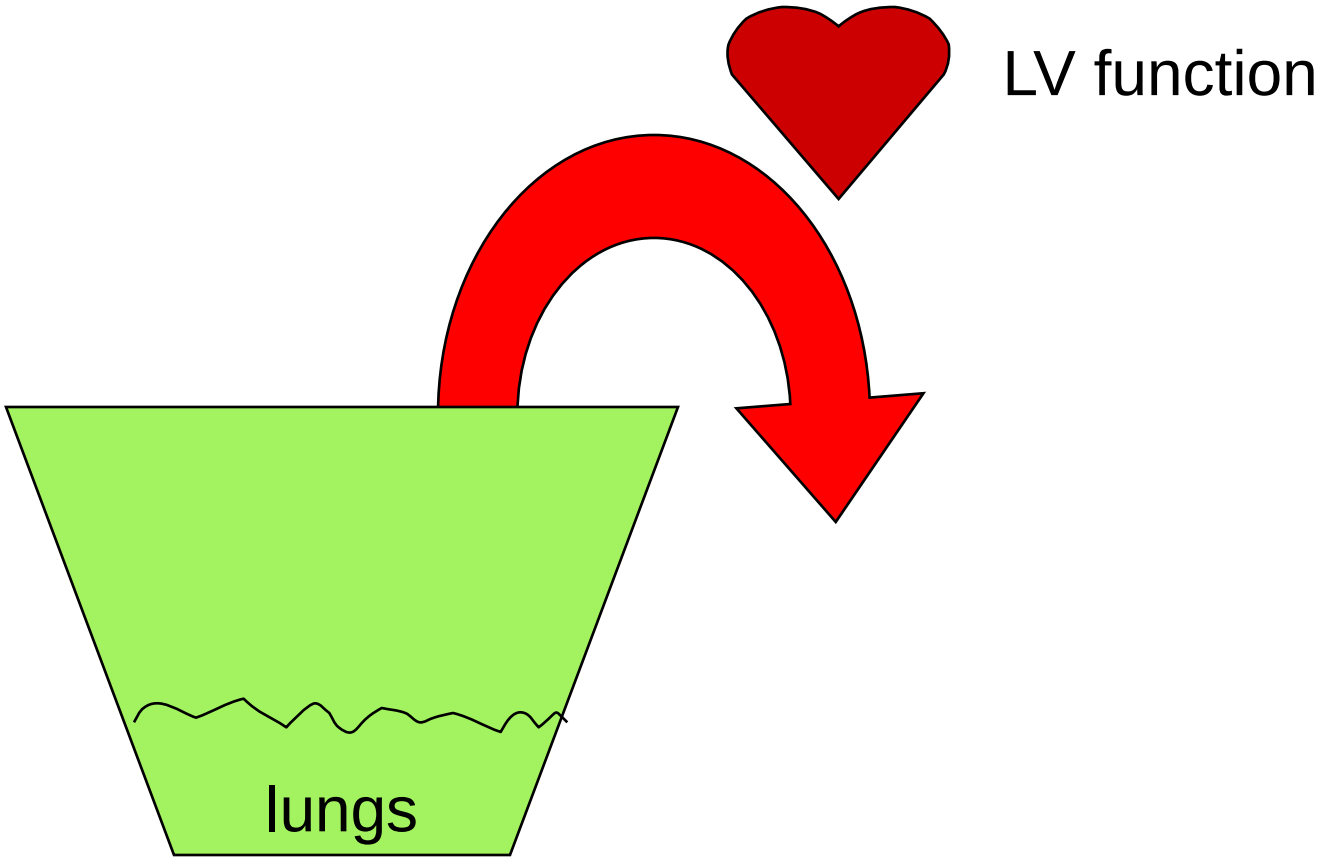
Pulmonary edema!!



2. increased afterload

PATHOPHYSIOLOGY



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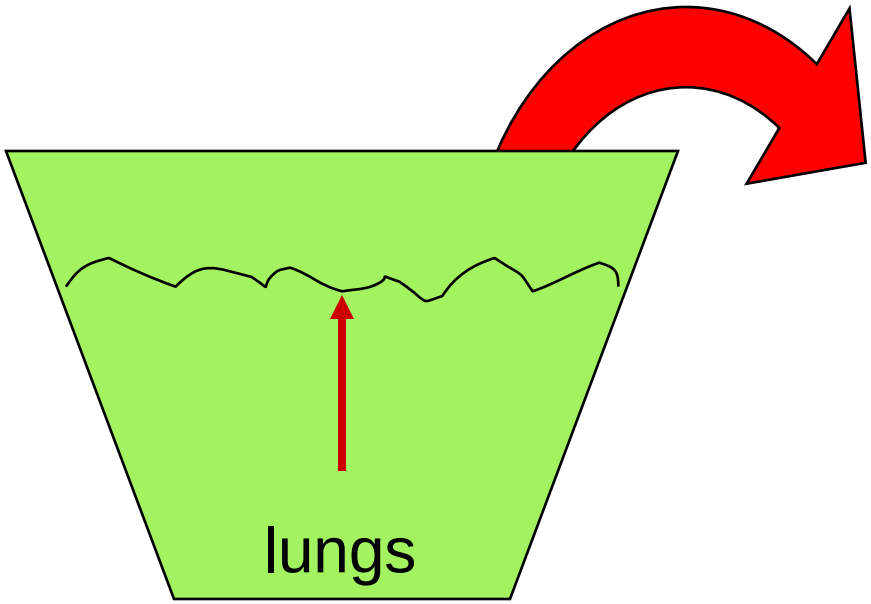
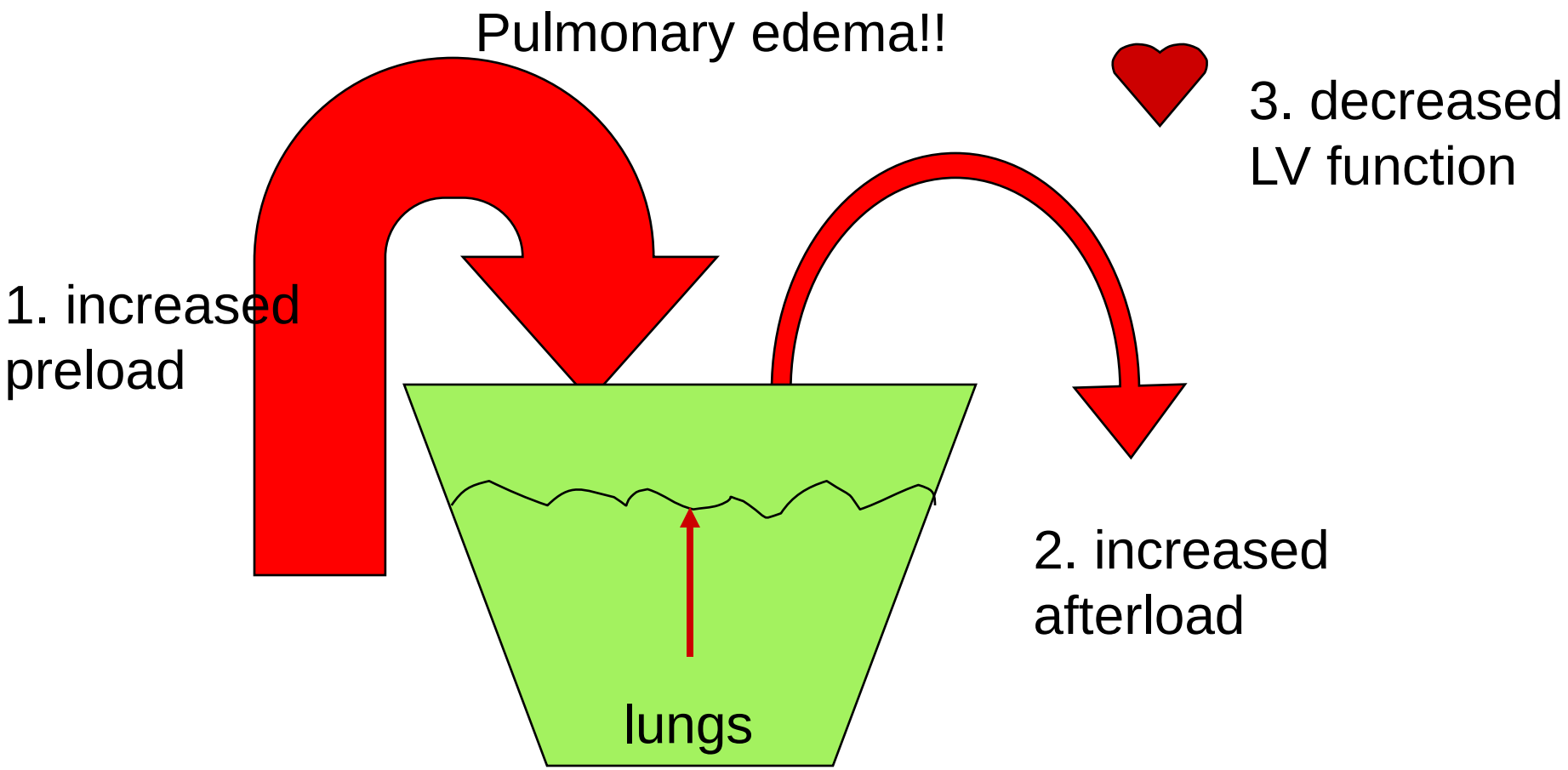


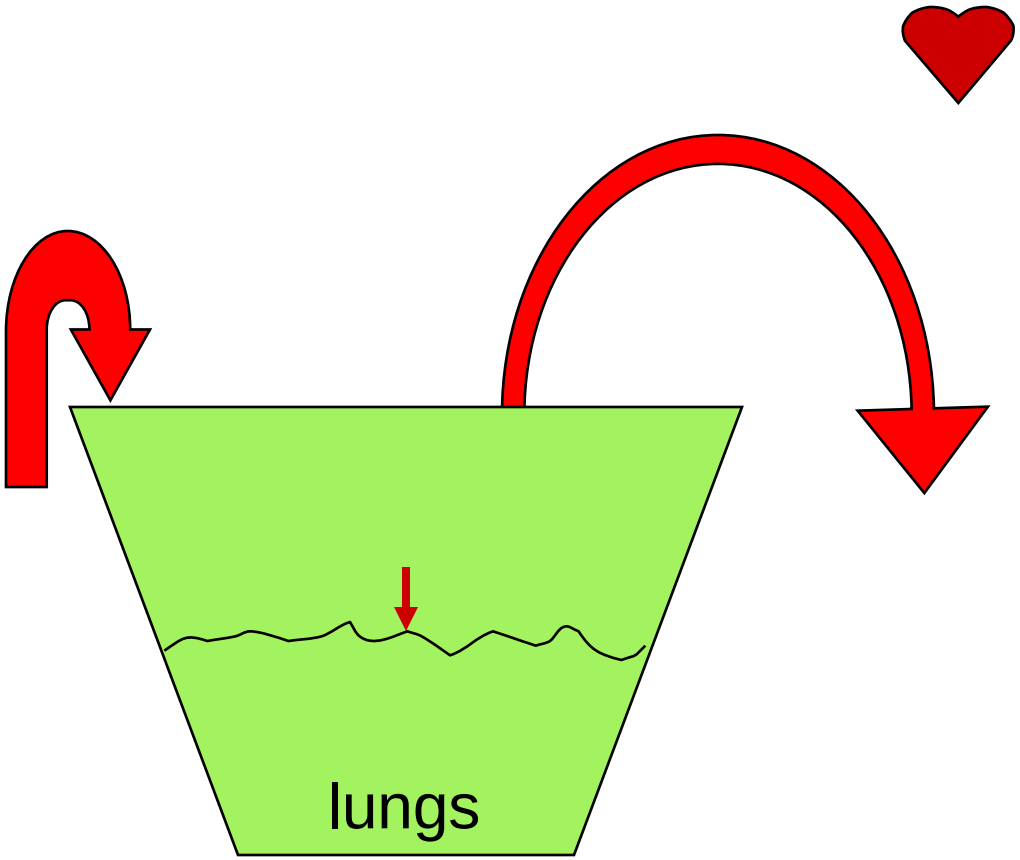
3. decreased LV function

PATHOPHYSIOLOGY



GOALS OF TREATMENT

1. decrease preload



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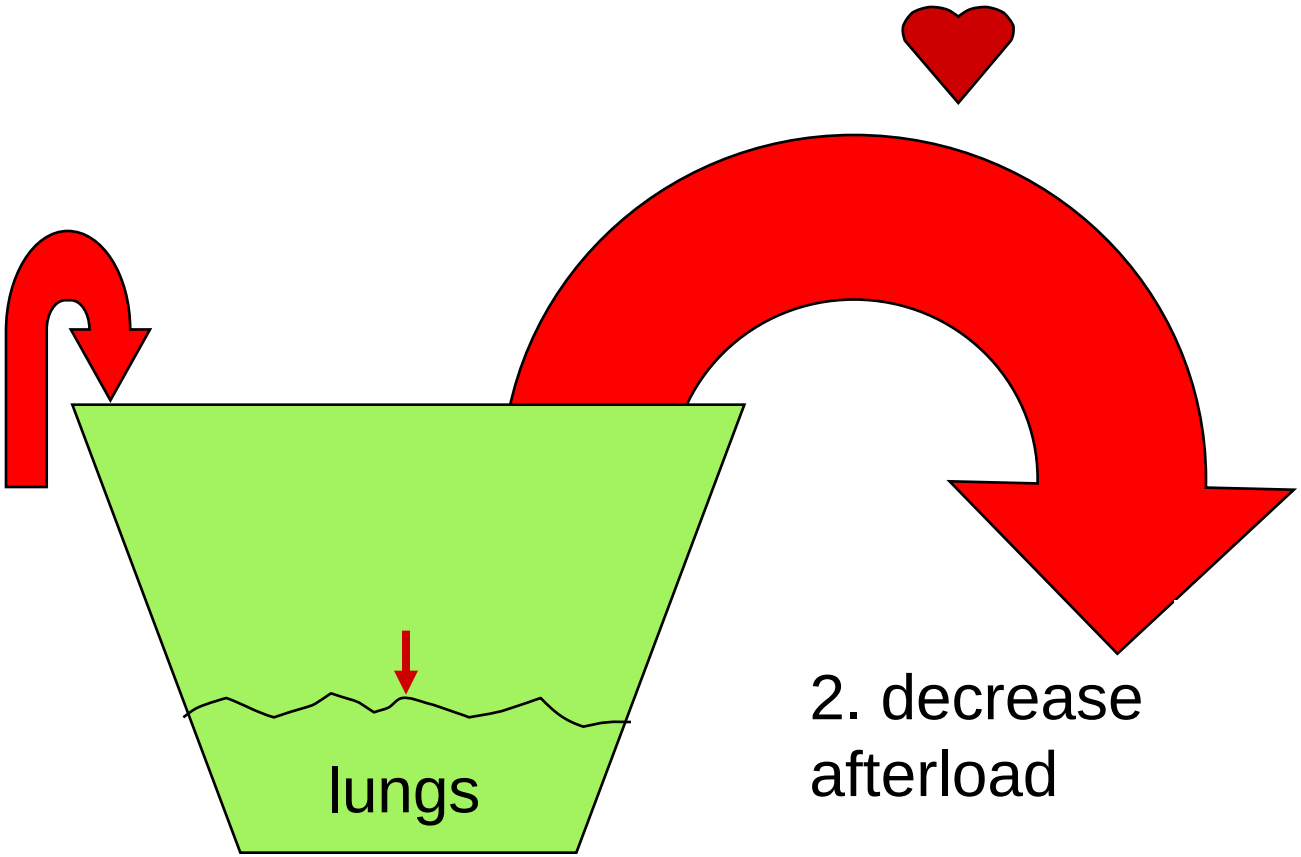

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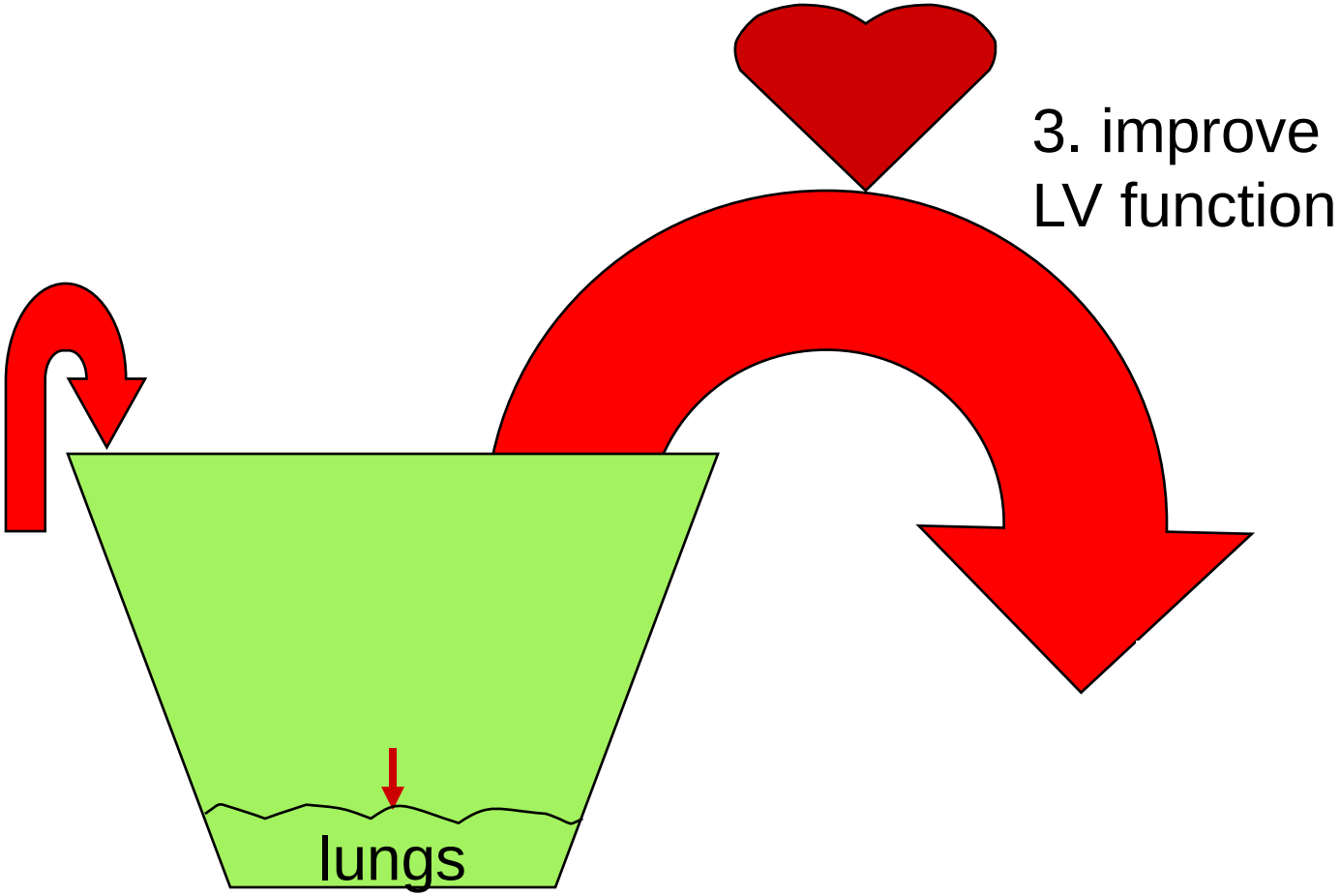

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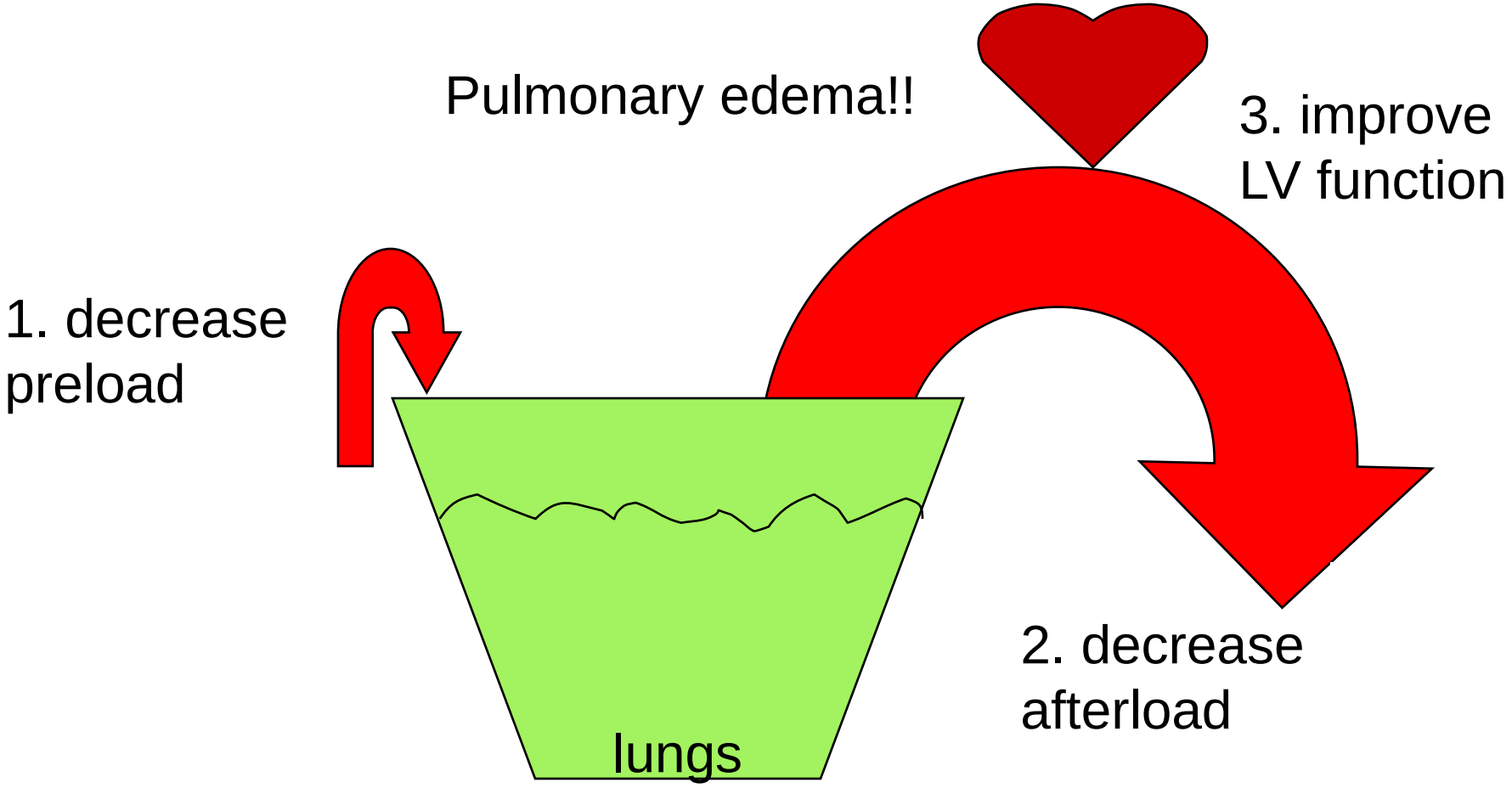

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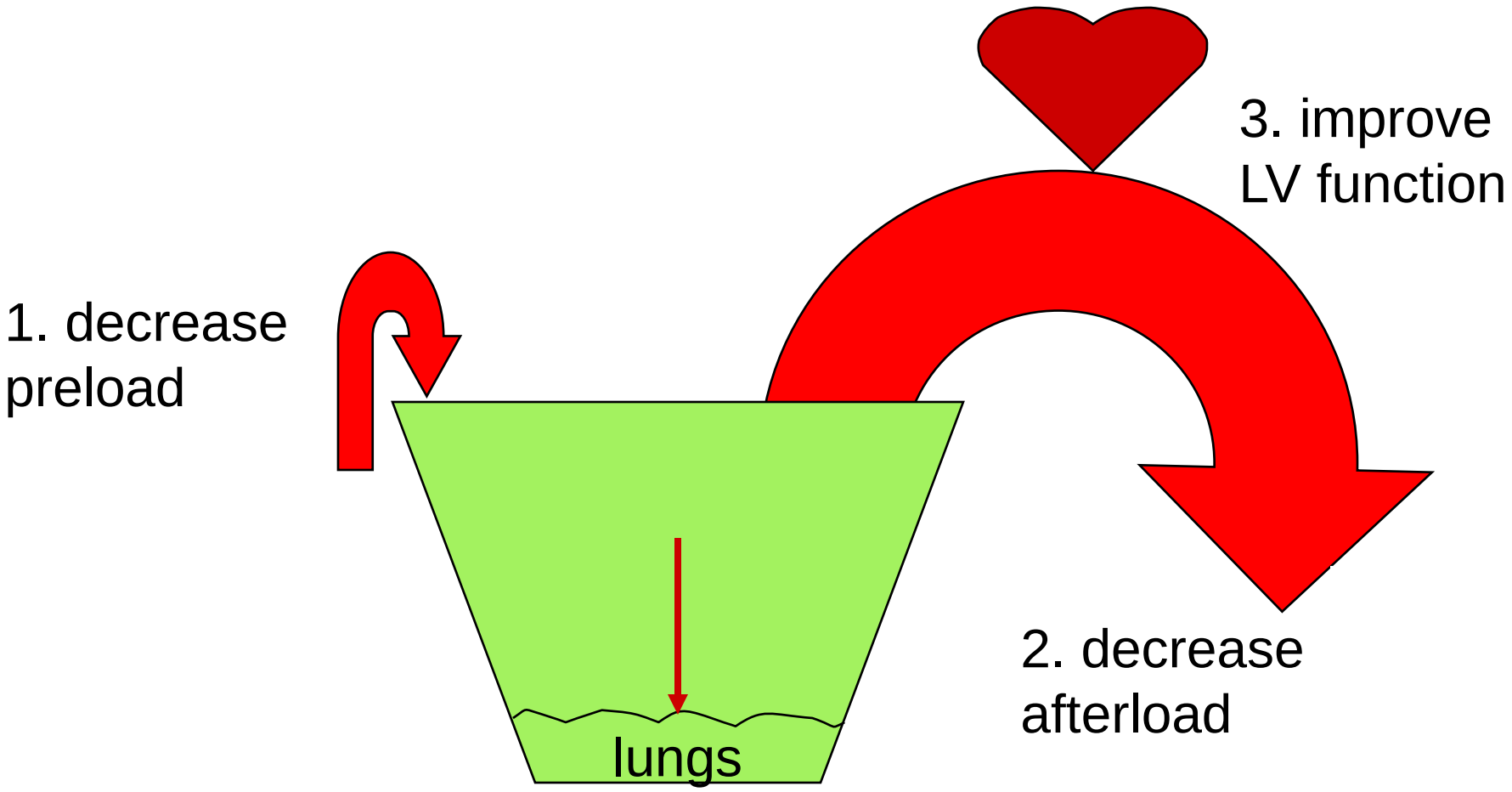

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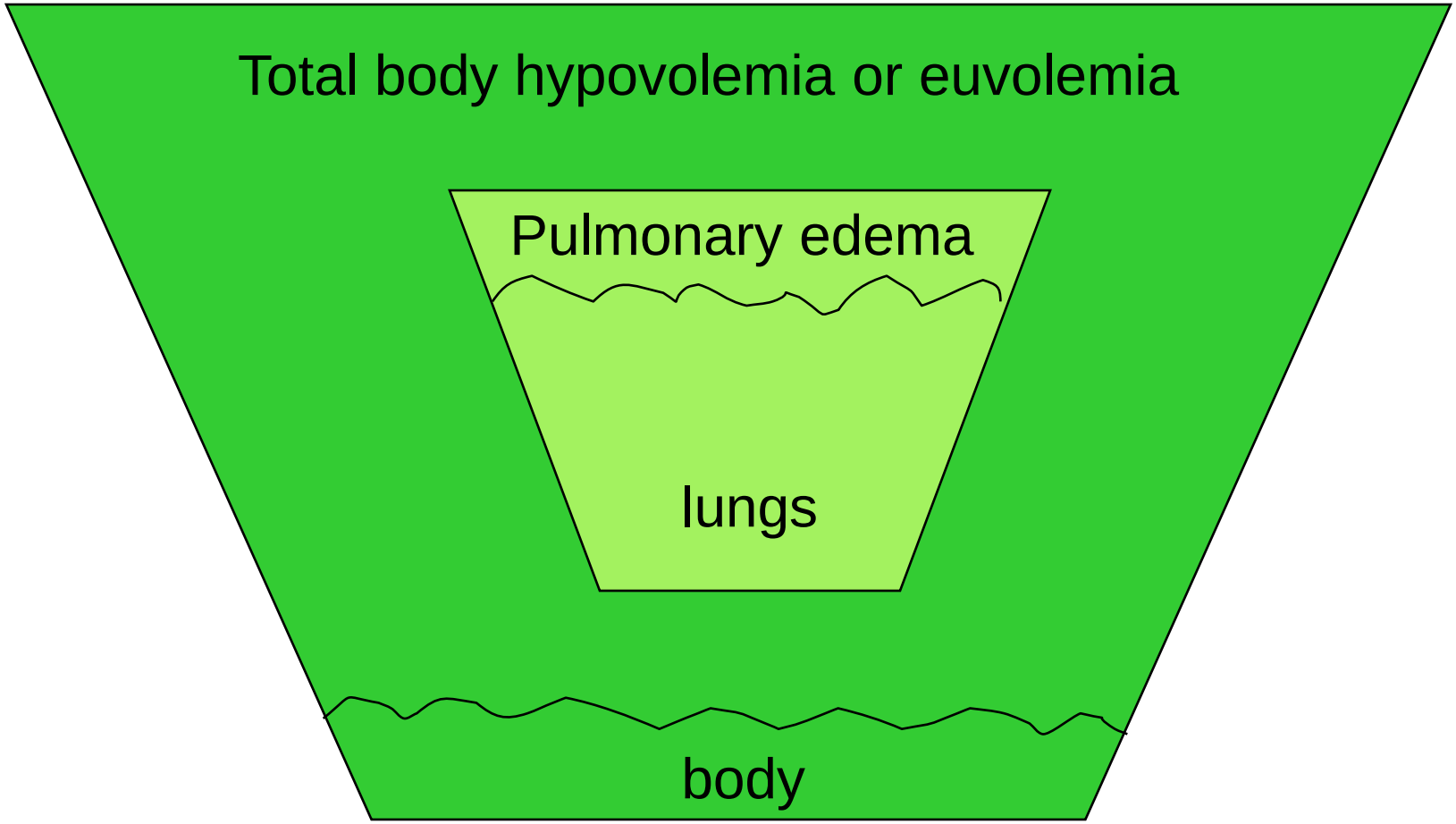

CARDIOGENIC PULMONARY EDEMA

- **<u>Note:</u>** more than 50% of patients with cardiogenic pulmonary edema are euvolemic!!
- Treatment should be based not necessarily on fluid *removal*, but on fluid *redistribution*.

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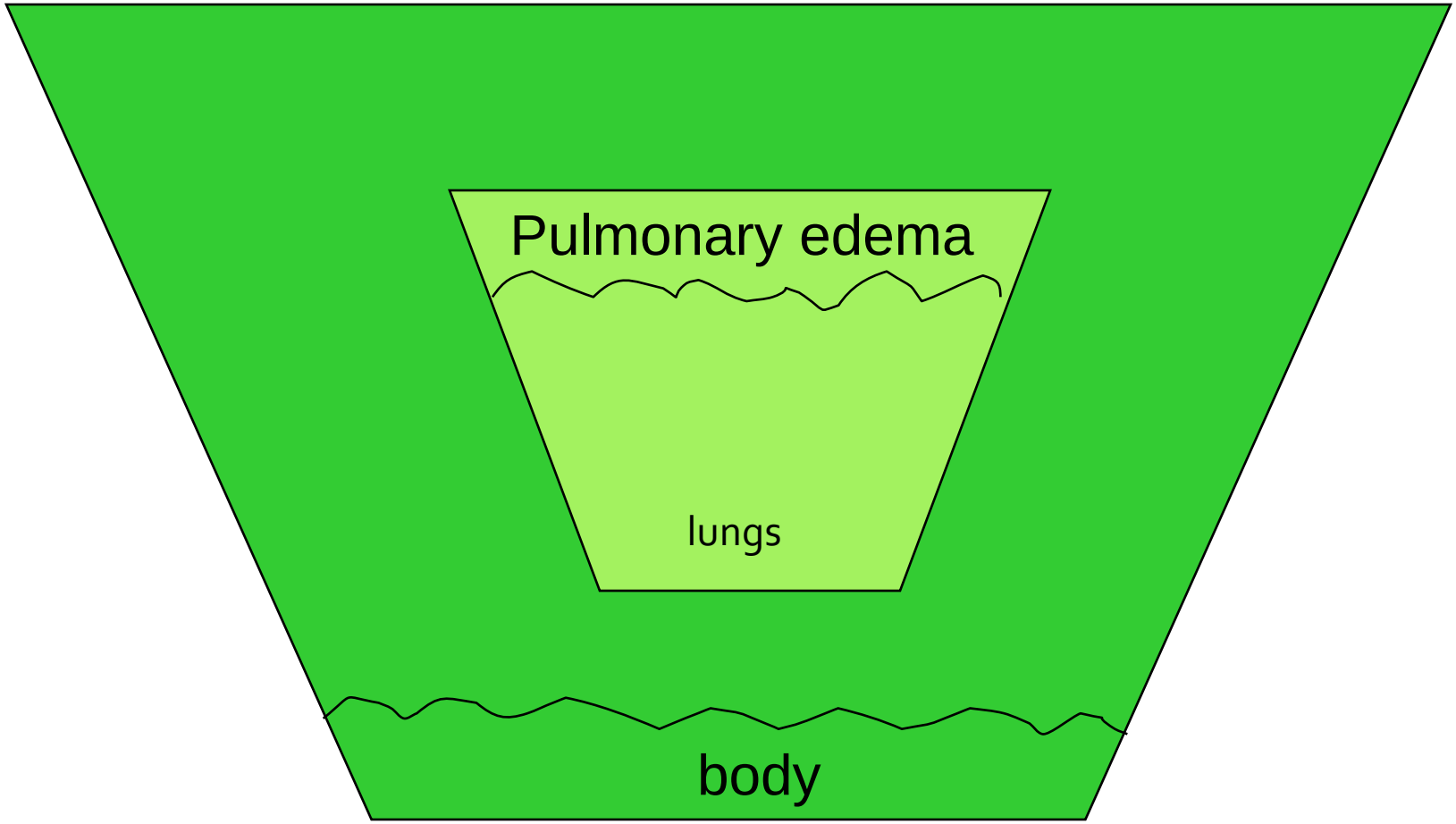

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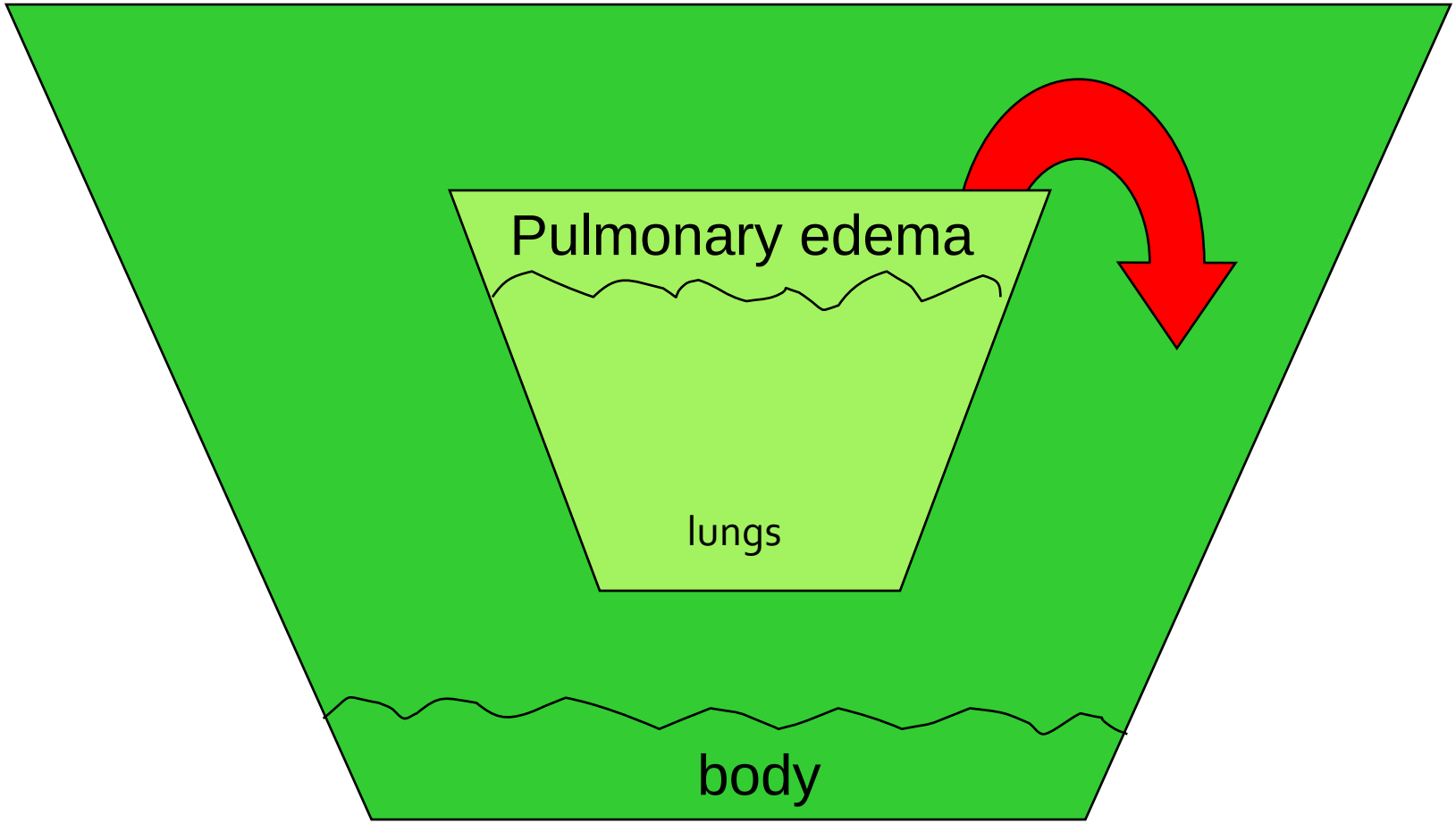

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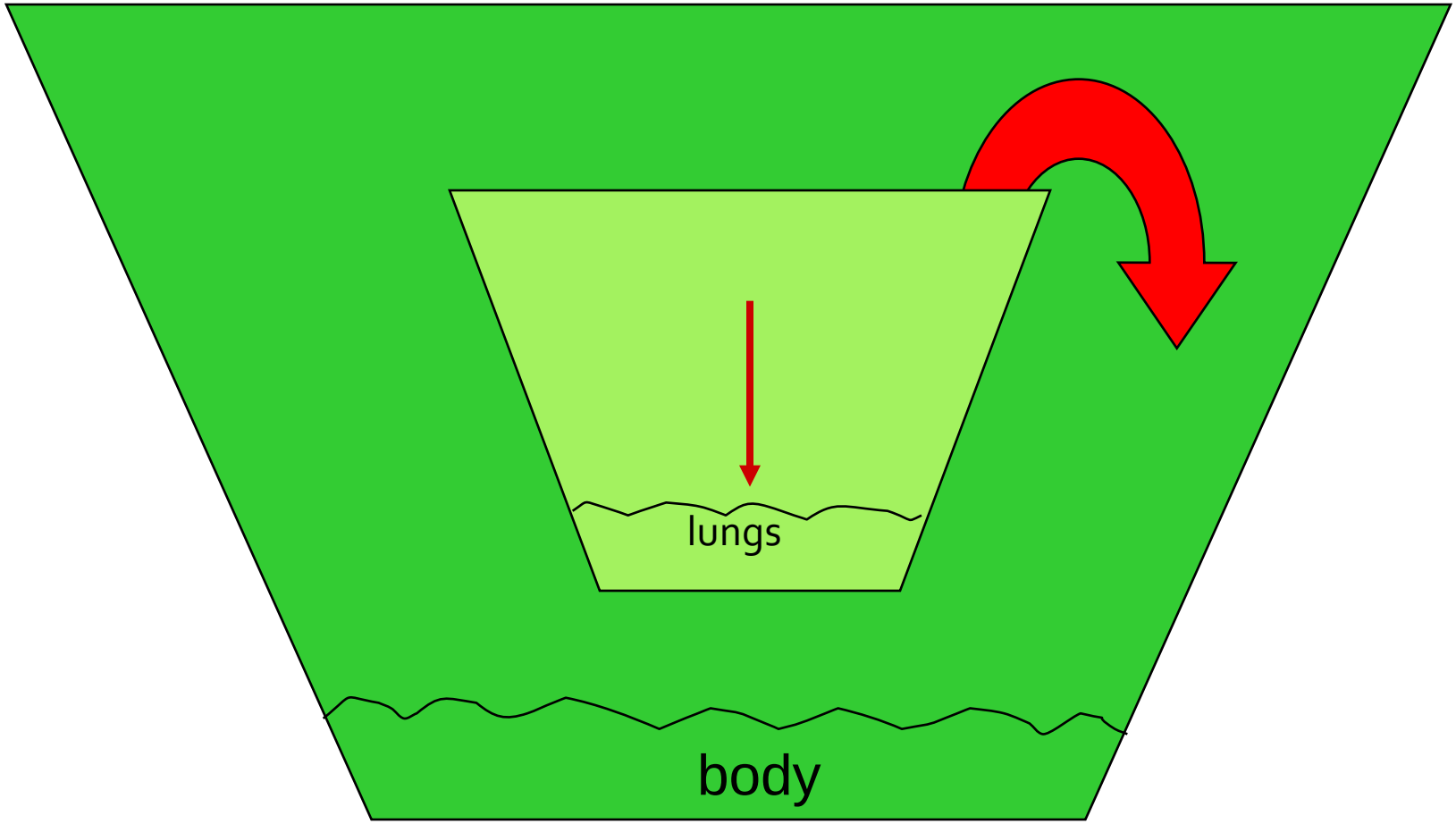

CARDIOGENIC PULMONARY EDEMA

- Causes
  - Excessive venous return (preload)
  - Excessive SVR (afterload)
  - LV dysfunction
    - disorders of contractility
    - disorders of rate and rhythm

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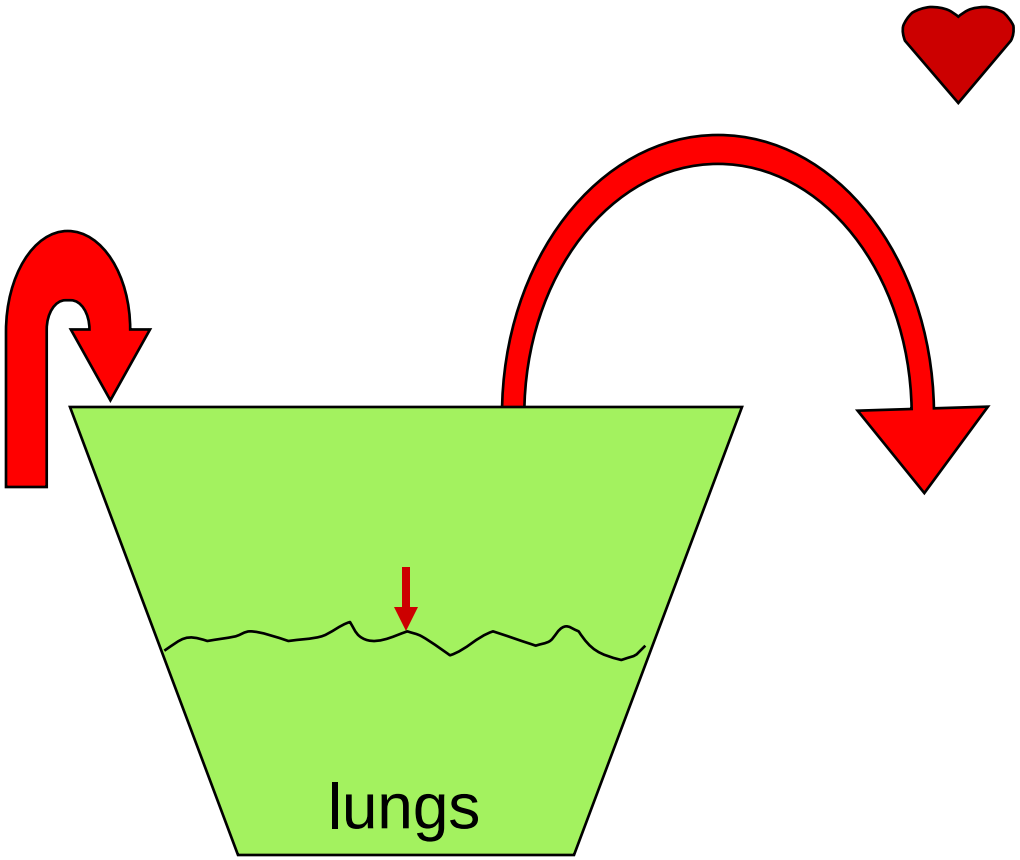

PRELOAD REDUCTION

- Traditional treatment
  - Morphine
  - Furosemide
  - Nitrates

MORPHINE

- Sacchetti, et al (*Am J Emerg Med*, 1999)
  - Odds ratios for intubation and ICU admission for pulmonary edema patients
  - Morphine — 5 : 1

MORPHINE

- Peacock WF, et al (*Emerg Med J*, 2008)
  - Morphine vs. no morphine for acute decompensated heart failure
    - independent predictor of mortality (OR 4.84)
    - increased need for mechanical ventilation, ICU admission, prolonged hospitalization

CONCLUSION — MORPHINE

Morphine has no role in the treatment of cardiogenic pulmonary edema!

FUROSEMIDE

Where's the debate??

FUROSEMIDE

- Preload reduction
  - Diuresis
  - Direct vasoactivity

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    - delayed effect: 30 – 120 minutes
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  - Diuresis
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    - delayed effect: 30 – 120 minutes
  - Direct vasoactivity
    - venodilation — little evidence
    - does this correlate with decreased preload?

FUROSEMIDE

- Pickkers P, et al (*Circulation*, 1997)
  - Direct vascular effects of furosemide on the human forearm vascular bed and dorsal hand vein
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  - Post-AMI CHF
  - Furosemide administered to 15 patients
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  - IV furosemide administration in acute CHF
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  - IV furosemide (1 mg/kg) administration in AMI patients with LV failure
  - Initial adverse hemodynamic effects
    - decreases in CO and SV during initial 90 minutes
  - Parameters returned to baseline over next 60 – 90 minutes

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FUROSEMIDE

- Francis, et al (*Ann Intern Med,* 1985)
  - Class III or IV CHF patients given IV furosemide
  - Early adverse hemodynamic effects
    - 20 minutes after administration
      - significant increase in HR, SVR (afterload)
      - significant decrease in SV
  - Gradual return to baseline with diuresis

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FUROSEMIDE

- Kraus, et al (*Chest,* 1990)
  - Effects of IV furosemide on PCWP over 1 hour in patients receiving nitrates and/or captopril
  - Furosemide alone or furosemide plus nitrates
    - increase in PCWP over initial 15 minutes
    - then decrease PCWP with diuresis
  - If *premedicated* with nitrates plus captopril
    - immediate and sustained decrease PCWP

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SUMMARY — FUROSEMIDE

- Decreases preload through diuresis
  - Delayed effect
- No consistent data regarding immediate direct preload reducing effect
- Initial adverse hemodynamic effects
  - Increased SVR
  - Decreased SV, CO

CONCLUSION — FUROSEMIDE

Furosemide should be considered a *third priority* in the treatment of cardiogenic pulmonary edema!

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NITROGLYCERIN

- Nitroglycerin vs. furosemide for preload reduction
  - Cotter, et al (*Lancet*, 1998)
  - Beltrame, et al (*J Card Fail*, 1998)
  - Kraus, et al (*Chest*, 1990)
  - Hoffman, et al (*Chest*, 1987)
  - Nelson, et al (*Lancet*, 1983)

NITROGLYCERIN

- Advantages
  - Rapid, reliable preload reduction
  - Moderate/high doses reduce SVR (afterload)
    - maintains or improves SV and CO
  - Multiple forms of administration — topical, SL, IV (be aggressive!)
  - Short half-life; especially important if prehospital misdiagnosis

NITROGLYCERIN

- Caution in the presence of…
  - Hypotension
  - Acute mitral regurgitation
  - Aortic stenosis
  - Pulmonary hypertension
  - Patients taking medications for E.D.

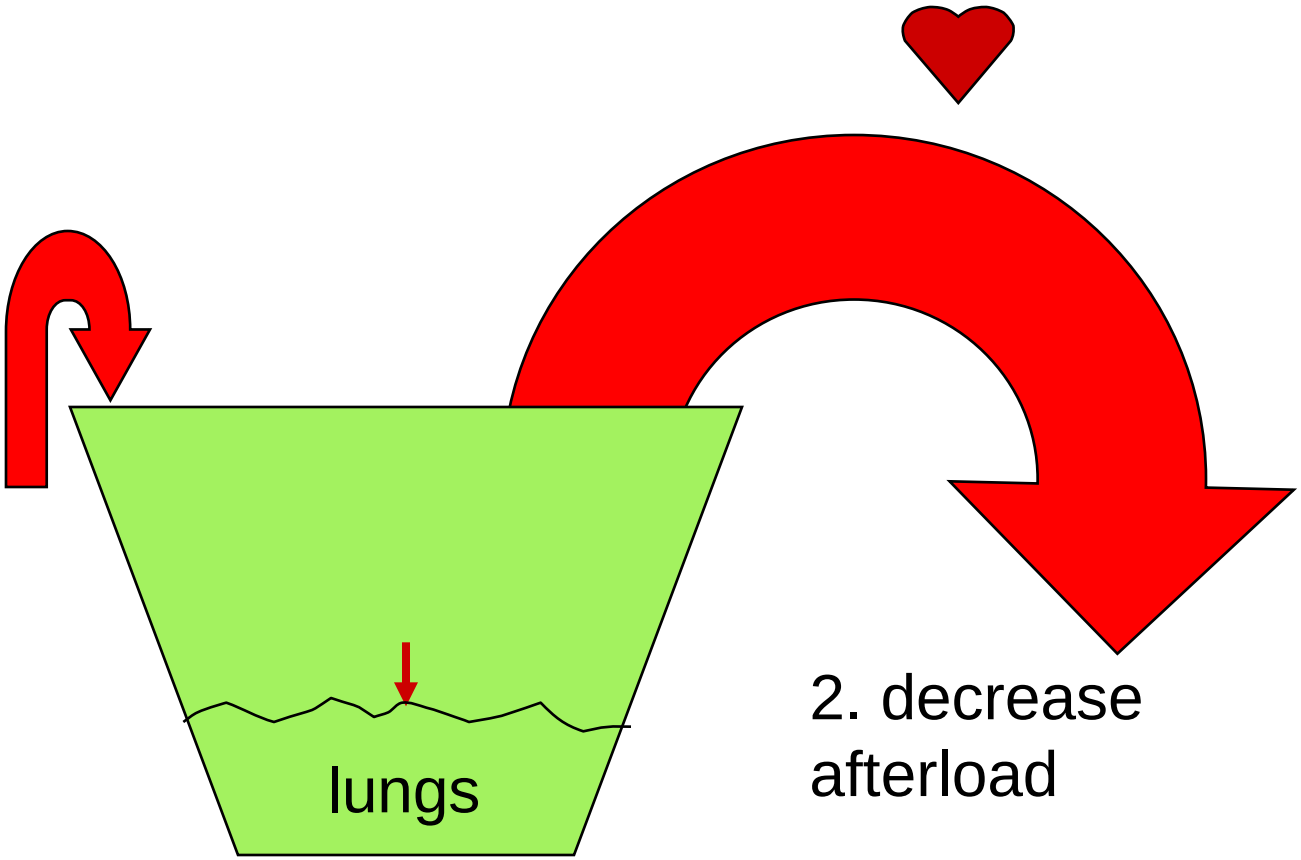
SUMMARY — NITROGLYCERIN

- Better than morphine or furosemide for preload reduction
- Safer than morphine or furosemide in prehospital setting
- SL nitroglycerin provides rapid and effective initiation of treatment
  - Followed by topical NTG if moderate CPE
  - Followed by IV NTG if severe CPE

CONCLUSION — NITROGLYCERIN

Nitroglycerin should be first-line prehospital and emergency department treatment for moderate CHF and pulmonary edema.

GOAL OF TREATMENT



AFTERLOAD REDUCTION

- Results in increased CO, restores renal blood flow
  - Nitroglycerin
    - SL and moderate/high dose IV
    - excellent single agent for simultaneous preload and afterload reduction
  - Nitroprusside
    - acute mitral regurgitation, severe hypertension
  - Hydralazine
  - ACE-inhibitors

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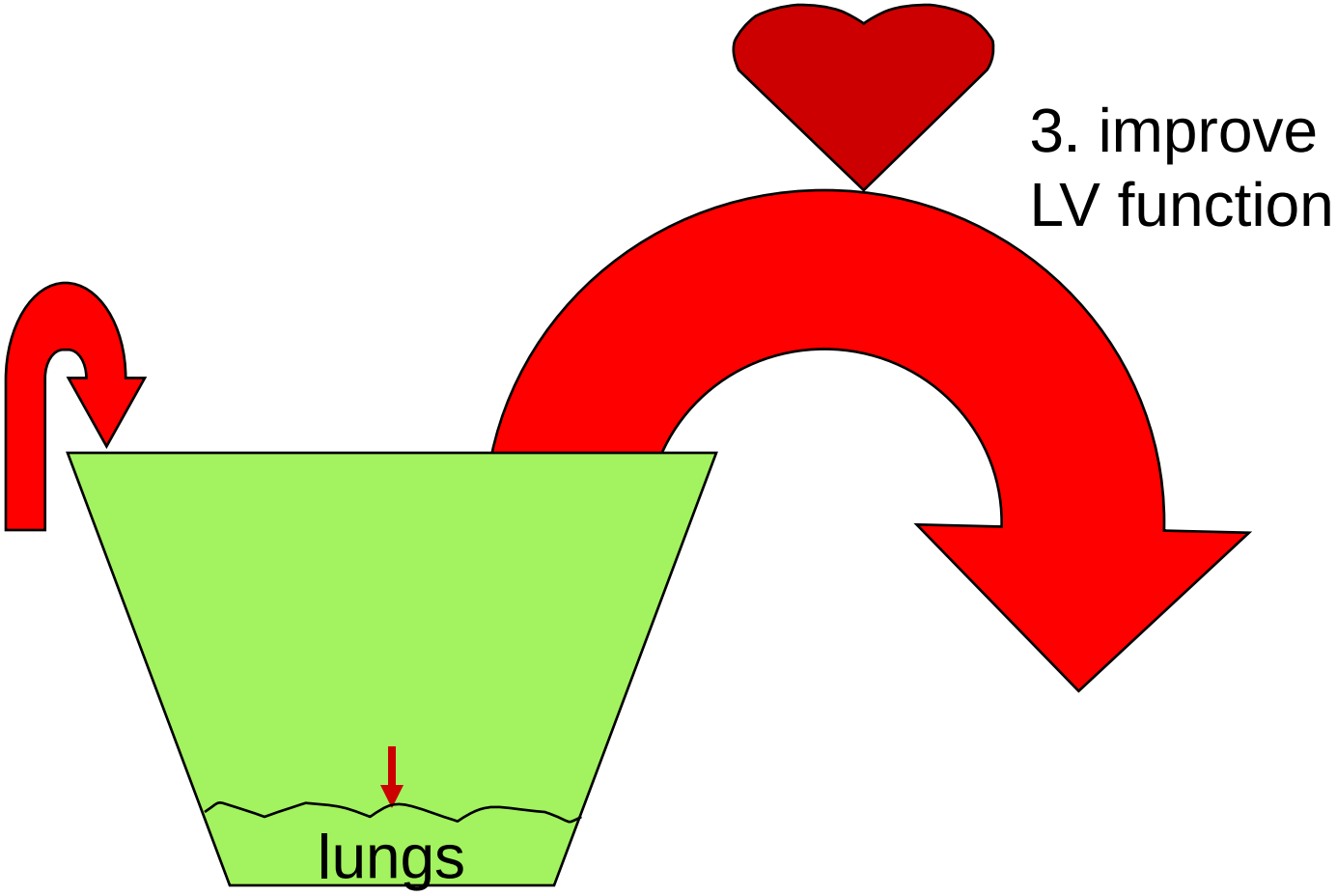
  - Hydralazine
  - ACE-inhibitors…for acute CPE!!

GOAL OF TREATMENT



INOTROPIC SUPPORT

- Choices
  - Catecholamines
    - dopamine
    - dobutamine
  - Phosphodiesterase inhibitors (milrinone)
  - IABP (bridging device before PTCA/CABG)

CONCLUSION — INOTROPIC SUPPORT

Use what you are comfortable with.

Consider use of milrinone for pulmonary edema patients unresponsive to DA/DBT.

NONINVASIVE POSITIVE PRESSURE VENTILATION

SUMMARY — NPPV

- Noninvasive positive pressure ventilation is associated with:
  - Decreased work of breathing
  - Improved O_2 and CO_2 exchange
  - Improved preload, afterload, and CO
  - Reduced need for intubation, ICU LOS
  - Reduced hospital costs

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  - Reduced need for intubation, ICU LOS
  - Reduced hospital costs
  - **Reduced mortality (?)**

SUMMARY — NPPV

- Noninvasive positive pressure ventilation is associated with:
  - NNT to save a life: 13
  - NNT to prevent an intubation: 8
  - No apparent harm

CONCLUSION — NPPV

NPPV is an effective method of providing airway support and averting intubation in some patients (but it *must be used early!!)*

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SUMMARY

- Nitroglycerin — first-line agent
  - IV nitroglycerin is excellent single-agent
- ACE-inhibitors or HIGH-dose NTG — second-line agent
  - In addition to or instead of nitroglycerin
- Furosemide — third-line agent
  - After preload and afterload reduction

SUMMARY

- Morphine???

SUMMARY

- Morphine — just say “no!”
  - Preload reduction — NTG more effective

SUMMARY

- Inotropic support
  - No data favors any specific agent…
  - Use what you are comfortable with!
- Noninvasive vent. — consider *early* use
  - Decreased intubations, ICU length of stay, and hospital costs

Case

- 65 yo. man presents with dyspnea
  - Developed over the past 6 hours
  - History of hypertension, tobacco use
  - Diaphoretic, normal mental status
  - Afebrile, HR 110, BP 180/110, RR 26, pox 88%
  - Lungs—crackles, JVD

Case

- 100% NRB—pulse ox 93%
- Monitor, IV, ECG
  - Sinus tachycardia, no acute ST/T-wave abnls.

Case

What is the optimal treatment in the first 5-10 minutes??

1. noninvasive ventilation
2. SL NTG q5 min (x 3-4 doses)
3. 1.25 mg IV enalapril or 25 mg SL captopril (OR rapid titration of high dose IV NTG)
4. furosemide AFTER above given

REMEMBER…

- More than 50% of patients with cardiogenic pulmonary edema are euvolemic!!
- Treatment should be based not necessarily on fluid *removal,* but on fluid *redistribution*.